

First Cut Sequence Diagram

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title First Cut Sequence Diagram: Memesan Barang (B2C)

skinparam sequenceMessageAlign center

skinparam style strictuml

actor "Customer" as Act

boundary "UI_Checkout" as UI

control "OrderHandler" as Ctrl

entity "MS_Product" as Prod

entity "MS_Customer" as Cust

entity "TR_Order" as Ord

entity "TR_DetailOrder" as Det

entity "TR_Payment" as Pay

Act → UI : checkout(cartData)

activate UI

UI → Ctrl : createOrder(cartData)

activate Ctrl

' Get Customer Info

Ctrl → Cust : getCustomerInfo(userID)

activate Cust

Cust → Ctrl : customerData

deactivate Cust

' Create Order Header

Ctrl → Ord : createNewOrder(customerData)

activate Ord

Ord → Ctrl : orderID
deactivate Ord

' Loop for items

loop for each item in cart

Ctrl → Prod : getPrice(productID)

activate Prod

Prod → Ctrl : price

deactivate Prod

Ctrl → Det : addLineItem(orderID, productID, qty, price)

activate Det

Det → Ctrl : success

deactivate Det

end

' Calculate Total

Ctrl → Ord : calculateTotal()

activate Ord

Ord → Ctrl : totalAmount

deactivate Ord

' Payment Process

Ctrl → Pay : createPaymentRecord(orderID, totalAmount)

activate Pay

Pay → Ctrl : paymentID

deactivate Pay

UI → Ctrl : processPayment(paymentMethod)

Ctrl → Pay : processTransaction(method)

activate Pay

```

alt Payment Success
    Pay → Ctrl : success
    Ctrl → Ord : updateOrderStatus("Paid")
    activate Ord
    deactivate Ord
    Ctrl → UI : displaySuccess()
else Payment Failed
    Pay → Ctrl : failed
    deactivate Pay
    Ctrl → Ord : updateOrderStatus("Failed")
    activate Ord
    deactivate Ord
    Ctrl → UI : displayError("Payment Failed")
end

deactivate Ctrl
deactivate UI
@enduml

```

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@startuml
title First Cut Sequence Diagram : Membuat Laporan Produksi
skinparam sequenceMessageAlign center
skinparam style strictuml

```

```

actor "Tim Produksi" as Act
boundary ":UI_Production" as UI
control ":ProductionReportingHandler" as Ctrl
entity ":TR_ProductionPlanning" as Plan
entity ":TR_ProductionControl" as Control
entity ":TR_ProductionResult" as Result
entity ":MS_MaterialWarehouse" as Stock

```

Act → UI : submitResult(jobOrderID, controlData, resultData)

activate UI

UI → Ctrl : submitProductionResult(jobOrderID, controlData, resultData)

activate Ctrl

' Validasi Job Order

Ctrl → Plan : validateJobOrder(jobOrderID)

activate Plan

Plan → Ctrl : valid

deactivate Plan

' Simpan Parameter Proses (Suhu/Viskositas)

Ctrl → Control : saveProcessParameters(jobOrderID, controlData)

activate Control

Control → Ctrl : saved

deactivate Control

' Simpan Hasil Output

Ctrl → Result : saveOutputData(jobOrderID, resultData)

activate Result

Result → Ctrl : resultID

deactivate Result

' Update Stok Barang Jadi

Ctrl → Stock : increaseStock(productID, resultData.qty, newBatchNo)

activate Stock

Stock → Ctrl : updated

deactivate Stock

```
' Tutup Job Order
Ctrl → Plan : updateJobStatus("Completed")
activate Plan
deactivate Plan
```

```
Ctrl → UI : showSuccess("Produksi Selesai")
deactivate Ctrl
deactivate UI
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```

```
@startuml
title First Cut Sequece Diagram : Manage Penerimaan Bahan Baku
skinparam sequenceMessageAlign center
skinparam style strictuml
```

```
actor "Staff Gudang" as Act
boundary ":UI_Receiving" as UI
control ":ReceivingHandler" as Ctrl
entity ":TR_Procurement" as PO
entity ":TR_PurchaseComplianceInspection" as QC
entity ":TR_DetailProcurement" as DetPO
entity ":MS_MaterialWarehouse" as Stock
entity ":MS_Warehouse" as Wh
```

```
Act → UI : scanPO(poID)
activate UI
UI → Ctrl : getPODetails(poID)
activate Ctrl
Ctrl → PO : getPOHeader(poID)
activate PO
deactivate PO
```

Ctrl → UI : showPOData
deactivate Ctrl

Act → UI : inputReceiving(batchData)
UI → Ctrl : receiveMaterial(poID, batchData)
activate Ctrl

' Validasi QC
Ctrl → QC : validateQCStatus(poID)
activate QC
QC → Ctrl : qcStatus
deactivate QC

alt QC Passed
Ctrl → PO : setStatus("Received")
activate PO
deactivate PO

loop for each item received
Ctrl → DetPO : updateBatchInfo(itemID, batchNo, expDate)
activate DetPO
deactivate DetPO

Ctrl → Wh : selectWarehouseLocation()
activate Wh
Wh → Ctrl : locationID
deactivate Wh

Ctrl → Stock : increaseStock(materialID, qty, batchNo, locationID)
activate Stock
deactivate Stock

```

end

    Ctrl → UI : showSuccess("Stok Masuk")
else QC Failed / Pending
    Ctrl → UI : showError("QC Belum Lolos")
end

deactivate Ctrl
deactivate UI
@enduml

```

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@startuml
title First Cut Sequence Diagram : Mengelola Pesanan (Omnichannel)
skinparam sequenceMessageAlign center
skinparam style strictuml

actor "System Ecommerce" as Ext
boundary ":API_Gateway" as API
control ":OmnichannelHandler" as Ctrl
entity ":MS_Product" as Prod
entity ":TR_Order" as Ord
entity ":TR_DetailOrder" as Det
entity ":MS_MaterialWarehouse" as Stock

Ext → API : pushOrder(jsonOrder)
activate API
API → Ctrl : syncOrdersFromMarketplace(jsonOrder)
activate Ctrl

' Cek Ketersediaan Stok

```

Ctrl → Stock : checkStock(sku, qty)

activate Stock

Stock → Ctrl : available

deactivate Stock

alt Stock Available

Ctrl → Ord : createOrderFromAPI(jsonOrder)

activate Ord

Ord → Ctrl : orderID

deactivate Ord

loop for items

Ctrl → Det : addItems(orderID, itemData)

activate Det

deactivate Det

end

' Reserve Stok (Soft Allocate)

Ctrl → Stock : reserveStock(sku, qty)

activate Stock

deactivate Stock

Ctrl → API : response(200, "Order Created")

else Stock Empty

Ctrl → API : response(409, "Out of Stock")

end

deactivate Ctrl

deactivate API

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title Sequence Diagram: Memantau Kinerja Bisnis (Dashboard)

skinparam sequenceMessageAlign center

skinparam style strictuml

actor "Pimpinan" as Act

boundary ":UI_Dashboard" as UI

control ":DashboardHandler" as Ctrl

entity ":TR_Order" as Ord

entity ":TR_ProductionResult" as Prod

entity ":MS_MaterialWarehouse" as Stock

entity ":TR_Procurement" as Proc

entity ":TR_Payment" as Pay

Act → UI : viewDashboard()

activate UI

UI → Ctrl : loadDashboardData()

activate Ctrl

par

Ctrl → Ord : getSalesSummary()

activate Ord

Ord → Ctrl : salesData

deactivate Ord

else

Ctrl → Prod : getProductionOutput()

activate Prod

Prod → Ctrl : productionData

deactivate Prod

else

Ctrl → Stock : getCriticalStock()

activate Stock

```

    Stock → Ctrl : stockData
    deactivate Stock
else
    Ctrl → Proc : getPendingApprovals()
    activate Proc
    Proc → Ctrl : approvalData
    deactivate Proc
else
    Ctrl → Pay : getCashFlowStatus()
    activate Pay
    Pay → Ctrl : financeData
    deactivate Pay
end parallel

Ctrl → UI : displayCharts(sales, prod, stock, finance)
deactivate Ctrl
deactivate UI
@enduml

```

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@startuml
title Sequence Diagram: Mengelola Laporan Keuangan
skinparam sequenceMessageAlign center
skinparam style strictuml

```

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actor "Accounting" as Act
boundary ":UI_Finance" as UI
control ":FinanceHandler" as Ctrl
entity ":TR_Payment" as Income
entity ":TR_Procurement" as Expense
entity ":TR_Delivery" as ShippingCost

```

Act → UI : generateReport(startDate, endDate)

activate UI

UI → Ctrl : generateReport(startDate, endDate)

activate Ctrl

' Get Income

Ctrl → Income : getIncomeByPeriod(startDate, endDate)

activate Income

Income → Ctrl : totalIncome

deactivate Income

' Get Material Expenses

Ctrl → Expense : getExpensesByPeriod(startDate, endDate)

activate Expense

Expense → Ctrl : materialCost

deactivate Expense

' Get Shipping Cost

Ctrl → ShippingCost : getShippingCost(startDate, endDate)

activate ShippingCost

ShippingCost → Ctrl : deliveryCost

deactivate ShippingCost

' Calculate Profit

Ctrl → Ctrl : calculateProfit(totalIncome, materialCost, deliveryCost)

Ctrl → UI : showFinancialReport(netProfit)

deactivate Ctrl

deactivate UI

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